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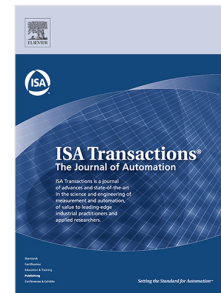
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Adaptive Trajectory Tracking Control for Remotely Operated Vehicles

Considering Thruster Dynamics and Saturation Constraints

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Abstract: This paper discusses the problem of adaptive trajectory tracking control for remotely operated vehicles (ROVs). Considering thruster dynamics, a third-order state space equation is used to describe the dynamic model of ROVs. For the problem of unknown dynamics and partially known input gain, an adaptive sliding mode control design scheme based on RBF neural networks is developed using a backstepping design technique. Because of the saturation constraints of the thrusters, a first-order auxiliary state system is applied, and subsequently, a saturation factor is constructed for designing adaptive laws to ensure the stability of the adaptive trajectory tracking system when the thrusters are saturated. The proposed controller guaranteed that trajectory tracking errors are uniformly ultimately bounded (UUB). Finally, the effectiveness of the proposed controller is verified by simulations.

Keywords: remotely operated vehicle, saturation constraints, trajectory tracking, adaptive control, thruster dynamics

1. Introduction

ROVs are used to perform various tasks in ocean engineering because of their efficiency [1–2]. Some tasks require high-level trajectory tracking control in terms of agility and precision [3–6]. However, such control is complicated because operators must monitor and control multiple parameters. If the trajectory could be tracked automatically, ROV operation would be greatly simplified [7].

Due to the strong nonlinearity of ROV motion system and external disturbances, various trajectory tracking control algorithms based on adaptive learning have been widely used in the past [8–13]. In [14], considering the highly nonlinear and uncertain of ROV systems, a sliding mode control method based on genetic algorithms and fuzzy system was discussed. In [15], an adaptive neural networks sliding mode method has been developed for the visual servo control of ROVs. In [16], the trajectory tracking control in the horizontal motion plane was studied, and an adaptive controller based on reinforcement learning was proposed. In [17], an independent adaptive fuzzy control method was given for the underwater vehicle motion control with uncertainties. Compared with traditional PID controller,

adaptive controllers based on neural networks have better tracking performance [18]. Adaptive control is considered effective for ROV trajectory tracking control. However, existing adaptive control methods have rarely focused on trajectory tracking with unknown input gain. In [19], considering the unknown dynamics and unknown input gain, two independent neural networks were used for on-line identification, respectively; subsequently, an adaptive trajectory tracking control method was proposed. Although this approach guarantees stability, selecting the parameters of the two neural networks is difficult in practice, especially for higher-order systems.

ROVs are fundamentally driven by propeller thrusters. Most propeller thrusters are designed as an independent closed-loop speed control system, and the control signal of thruster is an analog voltage [20]. In existing adaptive controllers for ROV trajectory tracking control, it is generally assumed that thrusters can respond to the desired thrust instantaneously without considering thruster dynamics [21]. In [22], considering the fault tolerant control problem, an adaptive neural network controller was proposed for underwater vehicles, and a linear model $F=BV$ was used to describe the relationship between the thruster control signal and thrust, where F is the force/moment acted on the underwater vehicles, V is the control signal, and B is the distribution matrix of thrusters. However, the ROV's power is usually provided by shore equipment through an umbilical cable [23]. Owing to the limitations of the cable diameter and the watertight connectors, the current to the thrusters must be reduced [24–26]. Moreover, current limiting helps improve the service life of thrusters. Therefore, the thrust output has a certain delay after the control signal is given. Considering thruster dynamics in the adaptive controller design is effective at improving the control performance of trajectory tracking, as well as obtaining the thruster's control signals directly through a control law.

In controller design, thrusters' saturation constraint is another issue that must be considered [27,28]. In particular, if the saturation constraint is neglected in an adaptive controller design, the system becomes unstable when the thrusters work in the saturation region for a continuous time [29]. In

traditional ROV control, the force or torque of each degree of freedom is first designed by the controller, following which the thrust allocation method is applied to calculate each thruster's desired thrust [30,31]. Some optimization methods can be used to ensure that the thruster is not saturated [32]. This approach is feasible but troublesome. At present, few studies have focused on adaptive controllers for ROV systems with saturation constraints. However, various algorithms have been proposed for other nonlinear systems [33–35]. Among them, the auxiliary state systems were commonly used to compensate for the actuator's saturation constraints [36,37]. In these auxiliary state systems, the input is the difference between the desired control law and the saturation boundary value. Based on the auxiliary state, a new state is constructed, and subsequently, the controller is designed to ensure that this new state is convergent or uniformly ultimately bounded (UUB). However, a prerequisite may be introduced for stability analysis. That is, the auxiliary state is bounded [36,37]. This is obviously an unreasonable assumption. The auxiliary state relies on the designed control law. Before controller design, its boundedness cannot be guaranteed.

Thus, this study proposes a novel adaptive trajectory tracking controller for ROV with which trajectory tracking errors are UUB. In comparison with the conventional ROV trajectory tracking control algorithm, the main innovations are as follows

- 1) Unlike most previous trajectory tracking controllers for ROVs, the thruster dynamics are considered in the proposed controller, and a third-order state space equation is constructed to describe the dynamic model of ROVs. In this dynamic model, the input is the control signal of the thrusters, and therefore, control signals could be designed directly by a controller.
- 2) Considering ROVs' trajectory tracking system with unknown dynamics and partially known input gain, an adaptive sliding mode controller is designed using the backstepping design technique, in which only the unknown dynamics are identified using RBF neural networks.
- 3) Considering the thruster saturation constraints, a saturation factor is constructed based on

first-order auxiliary state system, and a novel adaptive law combined with saturation factor is proposed. Unlike in previous auxiliary-state-system-based methods, the system stability is proved without assuming that the auxiliary state is bounded in advance.

The remainder of this paper is organized as follows. Section 2 summarizes the problem of ROV trajectory tracking control with thruster dynamics and saturation constraints. Section 3 presents the design process of the adaptive sliding mode controller based on the backstepping technique. Section 4 gives the simulation results. And then the conclusions are addressed in Section 5.

2. Problem Statement

2.1 Dynamic model for ROVs

Usually, the dynamic model of ROVs can be expressed by the following mathematical equation as [38]:

$$\begin{aligned} \dot{\eta} &= J(\eta)v \\ M\dot{v} + C(v)v + D(v)v + G(\eta) + \Gamma_d &= \bar{B}\Gamma(\omega) \end{aligned} \quad (1)$$

where η denotes the earth-fixed vector of position and orientation; v is the body-fixed vector of velocity and angular velocity; $J(\eta)$ is the rotational transformation matrix from the body-fixed frame to the earth-fixed frame; $\Gamma(\omega)$ is the thrust vector; $\bar{B}$ is the thrusters distribution matrix; M is the inertia matrix including extra mass; $C(v)$ is the centripetal and Coriolis forces matrix; $D(v)$ is the hydrodynamic damping term; $G(\eta)$ is the combined gravitational and buoyancy forces vector; Γ_d is external disturbances; and ω is the propeller speed vector.

Remark 1. In this paper, it is assumed that the values of η , v and ω can be measured directly. It is clear that the hypothesis is practical. For η , it can be measured by ultra-short baselines (USBL), electronic compass and depth gauge. With regard to v , it can be measured by DVL for velocity and IMU for angular velocity. For ω , it can be calculated by the frequency of the feedback pulse signal of the thruster.

Remark 2. Generally speaking, the thruster configuration will ensure that the ROV moves freely in the horizontal plane. Given a certain ROV, $\bar{B}$ is usually a constant matrix. In the vertical plane, the thruster configuration of different ROVs is different, but at least one thruster is guaranteed to enable the ROV to float and dive. For rolling and pitching, if it can not be controlled by thrusters directly, the items in (1) related to uncontrollable states can be assumed as external disturbances. This assumption is reasonable because the ROV is designed to ensure that the angles of pitch and roll are zero at rest. Therefore, v in (1) is only the controllable state vector, and $\bar{B}$ is a row full rank matrix.

2.2 Thruster dynamics

For a propeller thruster, the simplified thrust model of the i^{th} thruster can be expressed as in [39]:

$$\Gamma_i = a\omega_i |\omega_i|, \quad i = 1, \dots, n \quad (2)$$

where ω_i represents the propeller speed of the i^{th} thruster; a is a positive variable; and n is the thrusters number.

The control system of the thruster driven by DC motor is described by the dashed line frame shown in Fig. 1 [40], where U_i is the thruster control signal, e_ω is the control error of speed, K_p and K_I are the selected gains of the proportional–integral (PI) controller, K_1 is the amplification factor of the motor driver, K_2 is the amplification factor of the speed sensor, U is the armature voltage, i_d is the armature current, L_d is the inductance, and R_d is the resistance, $E = C_e \omega_i$ and C_e is the back EMF constant, J is the inertia moment, b is the viscous friction coefficient, T_p is the load torque, $T = C_m i$ is the electromagnetic torque, and C_m is the torque constant.

In Fig. 1, the DC motor is a second-order system, and a PI controller is used for closed-loop speed control. By choosing a proper PI controller gains, the DC motor can respond to the control signal quickly and accurately. But in actual ROV control, a first-order digital filter is often used to limit the DC motor current, where T_1 is the time constant.

It should be noted that Fig. 1 is just a simplified description of the thruster control system, and some

components are ignored. For example, the actual thruster includes gear reducer, magnetic coupling and so on. If all these factors are considered, it would increase the complexity of the controller design. Conversely, it is also inappropriate to overlook the thruster dynamics as traditional methods did. Considering the long control cycle of ROV systems, the fast-dynamic response of the DC motor, and the fact that a large time constant is usually chosen for the first-order digital filter, a first-order nonlinear state equation described by (3) is used in this study to describe the thruster dynamics.

$$\dot{\Gamma} = g(\Gamma) + \gamma V \quad (3)$$

where $V = [V_1, \dots, V_i, \dots, V_n]^T$ is the thruster control signal vector, $i = 1, \dots, n$, $g(\Gamma)$ is an unknown function vector, and $\gamma > 0$ is an unknown parameter.

By considering the thruster dynamics provided by (3), the dynamic model of ROVs is a third-order system, and the input of the dynamic model is the thruster's control signals.

2.3 Saturation constraints

In actual ROV systems, saturation constraints are unavoidable. The saturation constraints of the i^{th} thruster Γ_i and j^{th} control signal V_j are expressed as

$$\Gamma_i = \text{sat}(\tau_i) = \begin{cases} \tau_{i\max} & \tau_i \geq \tau_{i\max} \\ \tau_i & \tau_{i\min} < \tau_i < \tau_{i\max} \\ \tau_{i\min} & \tau_i \leq \tau_{i\min} \end{cases}, i = 1, \dots, n \quad (4)$$

$$V_j = \text{sat}(v_j) = \begin{cases} v_{j\max} & v_j \geq v_{j\max} \\ v_j & v_{j\min} < v_j < v_{j\max} \\ v_{j\min} & v_j \leq v_{j\min} \end{cases}, j = 1, \dots, n \quad (5)$$

where τ_i and v_i are given by the designed controller, $\tau_{i\max} > 0$ and $\tau_{i\min} < 0$ are the limit thrust value of forward and reverse of the i^{th} thruster, respectively, and $v_{j\max} > 0$ and $v_{j\min} < 0$ are the maximum and minimum control signal of the j^{th} thruster, respectively.

Remark 3. In the subsequent controller design, a backstepping design will be adopted, and τ_i is designed as a virtual control law. In some previous controllers, only the saturation constraints of the real control law are considered. However, we found that if the saturation constraint of the virtual

control law is ignored, the system would be unstable when the virtual control law is long time in a saturation state. Therefore, the saturation constraints of the real and virtual control laws should be both considered.

In (4), sharp corners exist when $\tau_i = \tau_{i\max}$ or $\tau_i = \tau_{i\min}$; therefore, a control law cannot be constructed directly based on the backstepping technique. A smooth function (6) as shown in Fig. 2 is used to approximate (4) [41].

$$h_i(\tau_i) = \begin{cases} \tau_{i\max} \frac{e^{\tau_i/\tau_{i\max}} - e^{-\tau_i/\tau_{i\max}}}{e^{\tau_i/\tau_{i\max}} + e^{-\tau_i/\tau_{i\max}}} & \tau_i \geq 0 \\ \tau_{i\min} \frac{e^{\tau_i/\tau_{i\min}} - e^{-\tau_i/\tau_{i\min}}}{e^{\tau_i/\tau_{i\min}} + e^{-\tau_i/\tau_{i\min}}} & \tau_i < 0 \end{cases}, i = 1, \dots, n \quad (6)$$

As can be seen from Fig. 2, there is a bounded function $d_i(\tau_i) = \text{sat}(\tau_i) - h(\tau_i)$, which represents the difference between the saturated function $\text{sat}(\tau_i)$ and the smooth functioning $h(\tau_i)$ [42].

Through the mean-value theorem, a positive variable σ ($0 < \sigma < 1$) exists, such that

$$h_i(\tau_i) = h_i(\tau_{0i}) + \left. \frac{dh_i(\tau_i)}{d\tau_i} \right|_{\tau_i=\tau_{0i}} (\tau_i - \tau_{0i}) \quad (7)$$

where $\tau_{0i} = \sigma \tau_i + (1 - \sigma) \tau_{0i}$, $\tau_{0i} \in [\tau_{i\min}, \tau_{i\max}]$.

Because $h_i(0) = 0$, by choosing $\tau_{0i} = 0$, (7) can be rephrased as $h_i(\tau_i) = h_{\sigma i} \tau_i$, where $h_{\sigma i} = dh_i(\tau_i)/d\tau_i|_{\tau_i=\tau_{0i}}$.

In a similar manner to (5), there is

$$\begin{aligned} \text{sat}(v_j) &= p_j(v_j) + q_j(v_j) \\ &= p_{\sigma j} v_j + q_j(v_j) \end{aligned} \quad (8)$$

where $q_j(v_j) = \text{sat}(v_j) - p_j(v_j)$,

$$p_j(v_j) = \begin{cases} v_{j\max} \frac{e^{v_j/v_{j\max}} - e^{-v_j/v_{j\max}}}{e^{v_j/v_{j\max}} + e^{-v_j/v_{j\max}}} & v_j \geq 0 \\ v_{j\min} \frac{e^{v_j/v_{j\min}} - e^{-v_j/v_{j\min}}}{e^{v_j/v_{j\min}} + e^{-v_j/v_{j\min}}} & v_j < 0 \end{cases}, j = 1, \dots, n \quad (9)$$

$$p_{\sigma j}(v_j) = \left. \frac{dp_j(v_j)}{dv_j} \right|_{v_j=v_{\sigma}} v_j \quad (10)$$

We define $X = [x_1^T, x_2^T, x_3^T]^T$, where $x_1 = \eta$, $x_2 = J(\eta)v$, and $x_3 = \tau$. Then, a third-order state space equation can be obtained according to (1) and (3):

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= f(x_1, x_2) + Bd + Bh_{\sigma}x_3 \\ \dot{x}_3 &= g(x_3) + \gamma q + \gamma p_{\sigma}v \end{aligned} \quad (11)$$

where, $d = [d_1(\tau_1), \dots, d_i(\tau_i), \dots, d_n(\tau_n)]^T$, $q = [q_1(v_1), \dots, q_j(v_j), \dots, q_n(v_n)]^T$, $h_{\sigma} = \text{diag}[h_{\sigma 1}(\tau_1), \dots, h_{\sigma i}(\tau_i), \dots, h_{\sigma n}(\tau_n)]^T$, $p_{\sigma} = \text{diag}[p_{\sigma 1}(v_1), \dots, p_{\sigma i}(v_i), \dots, p_{\sigma n}(v_n)]^T$, $v = [v_1, \dots, v_j, \dots, v_n]^T$, $f(x_1, x_2) = J'(\eta)v - J(\eta)M^{-1}(C(v)v + D(v)v + G(\eta) + \Gamma_d)$, and $B = J(\eta)M^{-1}\bar{B}$.

The control objective of this study is to develop a control law v based on adaptive sliding mode so that the desired trajectory x_{1d} can be tracked. To proceed with the controller design for (11), some assumptions as follows are used:

Assumption 1. x_{1d} , $\dot{x}_{1d}$ and $\ddot{x}_{1d}$ are all bounded.

Assumption 2. There is a positive constant π_1 to satisfy $0 \leq \pi_1 = \|1 - h_{\sigma}\| < 1$.

Assumption 3. In (4), γ is unknown; however, positive constants γ_0 and π_2 exist, such that $\gamma_0 > \gamma$ and $0 \leq \pi_2 = \|1 - \gamma p_{\sigma}/\gamma_0\| < 1$.

Remark 4. The value of $h_{\sigma i}$ varies with τ_{σ} . According to the definition, τ_{σ} is related to τ_i and not a specific value. As τ_i is the designed control law, $h_{\sigma i}$ is unknown that cannot be used in control law design. However, the range of values is certain. It satisfies $0 < h_{\sigma i} \leq 1$. Similarly, $0 < p_{\sigma j} \leq 1$. In other words, the input gains Bh_{σ} and γp_{σ} in (11) are partially known, and assumptions 2 and 3 may both hold.

3. Adaptive Trajectory Tracking Controller

3.1 RBF neural networks

Although the nonlinear functions $f(x_1, x_2)$ and $g(x_3)$ are unknown and the input gains are partially

known, in this study, only $f(x_1, x_2)$ and $g(x_3)$ are identified by RBF neural networks.

There are bounded weights W_f^* and W_g^* , such that [43]

$$F = f(x_1, x_2) + Bd = W_f^* \phi(x_1, x_2) + \varepsilon_f \quad (12)$$

$$G = g(x_3) + \gamma q = W_g^* \phi(x_3) + \varepsilon_g \quad (13)$$

where $\phi(\cdot)$ is an RBF, ε_f and ε_g are the optimal networks estimation errors.

The estimated outputs of F and G by using the RBF neural networks are

$$\hat{F} = \hat{W}_f \phi(x_1, x_2) \quad (14)$$

$$\hat{G} = \hat{W}_g \phi(x_3) \quad (15)$$

where $\hat{W}_f$ and $\hat{W}_g$ are the estimated weights.

Assumption 4 [44]. ε_f and ε_g are both bounded, and positive constants α_1 and α_2 exist, such that $\|\varepsilon_f\| \leq \alpha_1$ and $\|\varepsilon_g\| \leq \alpha_2$, respectively.

Assumption 5 [45]. The optimal weight matrices W_f^* and W_g^* are both upper-bounded.

3.2 Controller design

The backstepping design technique is used in this study.

Step 1. Define the trajectory tracking errors $e_1 = x_1 - x_{1d}$. Selecting a positive definite diagonal matrix C , the sliding mode surface is defined as

$$s = \dot{e}_1 + Ce_1 \quad (16)$$

By deriving s and according to (11) and (12), it can get

$$\dot{s} = W_f^* \phi(x_1, x_2) + \varepsilon_f + Cx_2 - \ddot{x}_{1d} - C\dot{x}_{1d} + Bh_\sigma x_3 \quad (17)$$

Considering the Lyapunov function to be

$$V_1 = \frac{1}{2} s^T s + \frac{1}{2\gamma_{11}} \tilde{\alpha}_1^2 + \frac{1-\pi_1}{2\gamma_{12}} \tilde{\beta}_1^2 + \frac{1}{2\gamma_{13}} \text{tr}(\tilde{W}_f^T \tilde{W}_f) \quad (18)$$

where $\tilde{\alpha}_1 = \alpha_1 - \hat{\alpha}_1$, $\hat{\alpha}_1$ is the estimate of α_1 ; $\tilde{\beta}_1 = \beta_1 - \hat{\beta}_1$, $\hat{\beta}_1$ is the estimate of $\beta_1 = 1/(1 - \pi_1)$;

$\tilde{W}_f = W_f^* - \hat{W}_f$; and γ_{11} , γ_{12} , and γ_{13} are the update step sizes of $\hat{\alpha}_1$, $\hat{\beta}_1$, and $\hat{W}_f$, respectively.

Differentiating V_1 and using (18) gives

$$\dot{V}_1 = s^T \left(\tilde{W}_f^* \phi(x_1, x_2) + \varepsilon_f + Cx_2 - \ddot{x}_{1d} - C\dot{x}_{1d} + Bh_\sigma x_3 \right) - \gamma_{11}^{-1} \tilde{\alpha}_1 \dot{\alpha}_1 - (1 - \pi_1) \gamma_{12}^{-1} \tilde{\beta}_1 \dot{\beta}_1 - \gamma_{13}^{-1} \text{tr} \left(\tilde{W}_f^T \dot{\tilde{W}}_f \right) \quad (19)$$

We chose the virtual control law x_{3d} as

$$x_{3d} = B^{-1} \left(-\hat{W}_f \phi(x_1, x_2) - Cx_2 + \ddot{x}_{1d} + C\dot{x}_{1d} - k_1 s - \hat{\alpha}_1 \text{sgn}(s) - \Lambda_1 \text{sgn}(s) \right) \quad (20)$$

where $\Lambda_1 = -\lambda_1 + \lambda_1 \hat{\beta}_1$ and $\lambda_1 = \|\hat{W}_f \phi(x_1, x_2) + Cx_2 - \ddot{x}_{1d} - C\dot{x}_{1d} + k_1 s\| + \|\hat{\alpha}_1\| + \kappa_1$, κ_1 is a positive constant.

The adaptive laws are expressed as (21)–(23), respectively, and the control gain k_1 is expressed as (24).

$$\dot{\hat{\alpha}}_1 = \gamma_{11} (1 - \mu_1) (\|s\| - m_{11} \hat{\alpha}_1) \quad (21)$$

$$\dot{\hat{\beta}}_1 = \gamma_{12} \left(1 - \frac{\mu_1}{1 + \lambda_1} \right) (\lambda_1 \|s\| - m_{12} \hat{\beta}_1) \quad (22)$$

$$\dot{\hat{W}}_f = \gamma_{13} (1 - \mu_1) (s \phi(x_1, x_2)^T - m_{13} \hat{W}_f) \quad (23)$$

$$k_1 = \frac{\mu_1^2}{m_{11} (1 - \mu_1)} + \frac{\mu_1^2}{m_{12} (1 - \mu_1)} + \frac{\mu_1^2}{m_{13} (1 - \mu_1)} \|\phi(x_1, x_2)\|^2 + \theta_1 \quad (24)$$

where m_{11} , m_{12} , m_{13} , and θ_1 are all positive constants, $\mu_1 = \|\xi_1\| / (\|\xi_1\| + 1)$ is a saturation factor, and ξ_1 is obtained using the following first-order auxiliary state system:

$$\dot{\xi}_1 = -L_1 \xi_1 + H_1 \Delta x_3 \quad (25)$$

where ξ_1 is an auxiliary state, $\Delta x_3 = x_{3d} - \text{sat}(x_3)$ is the saturation nonlinearity, and L_1 and H_1 are both positive constant matrices.

The estimation errors of the RBF neural networks are inevitable. They are difficult to converge completely to zero in reality. Especially at the initial time, trajectory tracking error is abrupt. The proportional items and symbolic function items are used in (20), so that the estimation error can be compensated for trajectory tracking control.

In (25), ξ_1 represents the filtered version of Δx_3 and remains zero as long as $\Delta x_3 = 0$. In this paper,

based on the auxiliary state, the saturation factor μ_1 is constructed and applied to the adaptive laws. It differs greatly from previous adaptive controllers. In [46–49], the adaptive law is usually chosen as $\gamma_{11}\|s\|$ or $\gamma_{11}(\|s\| - m_{11}\hat{\alpha}_1)$ for α_1 . These adaptive laws are found to possibly cause instability when the thruster is long time in the saturation region. The previous research found that this is due to $\hat{\alpha}_1$ must be updated according to $\|s\|$; however, when x_3 is in the saturation region, $\|s\|$ may not converge to zero. At this point, $\hat{\alpha}_1$ is updated with the wrong law continually, and intrinsically, it eventually makes the system unstable. Therefore, the saturation factor is used in this study. If the thrust is saturated, the adaptive laws will be reduced until they reach zero.

Substituting (20), (21) and (23) into (19) gives

$$\begin{aligned} \dot{V}_1 = & s^T \left[\tilde{W}_f^* \phi(x_1, x_2) - \hat{W}_f \phi(x_1, x_2) + \varepsilon_f - k_1 s - \hat{\alpha}_1 \operatorname{sgn}(s) - \Lambda_1 \operatorname{sgn}(s) - B(1 - h_\sigma) x_{3d} \right] - \\ & \gamma_{11}^{-1} \tilde{\alpha}_1 \dot{\hat{\alpha}}_1 - (1 - \pi_1) \gamma_{12}^{-1} \tilde{\beta}_1 \dot{\hat{\beta}}_1 - \gamma_{13}^{-1} \operatorname{tr} \left(\tilde{W}_f^T \dot{\hat{W}}_f \right) \\ \leq & -k_1 s^T s + \mu_1 \operatorname{tr} \left(\tilde{W}_f^T s \phi(x_1, x_2)^T \right) + m_{13} (1 - \mu_1) \operatorname{tr} \left(\tilde{W}_f^T \hat{W}_f \right) + \mu_1 \tilde{\alpha}_1 \|s\| + m_{11} (1 - \mu_1) \tilde{\alpha}_1 \hat{\alpha}_1 - \\ & \Lambda_1 \|s\| - s^T B(1 - h_\sigma) x_{3d} - (1 - \pi_1) \gamma_{12}^{-1} \tilde{\beta}_1 \dot{\hat{\beta}}_1 \end{aligned} \quad (26)$$

because

$$\begin{aligned} -s^T B(1 - h_\sigma) x_{3d} & \leq \|s\| \pi_1 \left[\left\| \tilde{W}_f \phi(x_1, x_2) + Cx_2 - \ddot{x}_{1d} - C\dot{x}_{1d} \right\| + \|\hat{\alpha}_1\| + \kappa_1 + \Lambda_1 - \kappa_1 \right] \\ & = \|s\| \pi_1 (\lambda_1 \hat{\beta}_1 - \kappa_1) \end{aligned} \quad (27)$$

Substituting (22) and (27) into (26) gives

$$\begin{aligned} \dot{V}_1 \leq & -k_1 s^T s - \kappa_1 \pi_1 \|s\| + \mu_1 \operatorname{tr} \left(\tilde{W}_f^T s \phi(x_1, x_2)^T \right) + m_{13} (1 - \mu_1) \operatorname{tr} \left(\tilde{W}_f^T \hat{W}_f \right) + \mu_1 \tilde{\alpha}_1 \|s\| + m_{11} (1 - \mu_1) \tilde{\alpha}_1 \hat{\alpha}_1 + \\ & (1 - \pi_1) \lambda_1 \tilde{\beta}_1 \|s\| - (1 - \pi_1) \gamma_{12}^{-1} \tilde{\beta}_1 \dot{\hat{\beta}}_1 \\ \leq & -k_1 s^T s + \mu_1 \operatorname{tr} \left(\tilde{W}_f^T s \phi(x_1, x_2)^T \right) + m_{13} (1 - \mu_1) \operatorname{tr} \left(\tilde{W}_f^T \hat{W}_f \right) + \mu_1 \tilde{\alpha}_1 \|s\| + m_{11} (1 - \mu_1) \tilde{\alpha}_1 \hat{\alpha}_1 + \\ & (1 - \pi_1) \frac{\lambda_1}{1 + \lambda_1} \mu_1 \tilde{\beta}_1 \|s\| + m_{12} (1 - \pi_1) \left(1 - \frac{\mu_1}{1 + \lambda_1} \right) \tilde{\beta}_1 \hat{\beta}_1 \end{aligned} \quad (28)$$

because

$$\begin{aligned} m_{11} (1 - \mu_1) \tilde{\alpha}_1 \hat{\alpha}_1 & = m_{11} (1 - \mu_1) \tilde{\alpha}_1 (\alpha_1 - \tilde{\alpha}_1) \\ & \leq -\frac{m_{11}}{2} (1 - \mu_1) \tilde{\alpha}_1^2 + \frac{m_{11}}{2} (1 - \mu_1) \alpha_1^2 \end{aligned} \quad (29)$$

$$\begin{aligned} m_{12}(1-\pi_1)\left(1-\frac{\mu_1}{1+\lambda_1}\right)\tilde{\beta}_1\hat{\beta}_1 &= m_{12}(1-\pi_1)\left(1-\frac{\mu_1}{1+\lambda_1}\right)\tilde{\beta}_1(\beta_1-\tilde{\beta}_1) \\ &\leq -\frac{m_{12}}{2}(1-\pi_1)\left(1-\frac{\mu_1}{1+\lambda_1}\right)\tilde{\beta}_1^2 + \frac{m_{12}}{2}(1-\pi_1)\left(1-\frac{\mu_1}{1+\lambda_1}\right)\beta_1^2 \end{aligned} \quad (30)$$

$$\begin{aligned} m_{13}(1-\mu_1)\text{tr}(\tilde{W}_f^T \hat{W}_f) &= m_{13}(1-\mu_1)\text{tr}(\tilde{W}_f^T (W_f^* - \tilde{W}_f)) \\ &\leq -\frac{m_{13}}{2}(1-\mu_1)\|\tilde{W}_f\|^2 + \frac{m_{13}}{2}(1-\mu_1)\|W_f^*\|^2 \end{aligned} \quad (31)$$

$$-\frac{m_{11}}{4}(1-\mu_1)\tilde{\alpha}_1^2 + \mu_1\tilde{\alpha}_1\|s\| \leq \frac{\mu_1^2}{m_{11}(1-\mu_1)}\|s\|^2 \quad (32)$$

$$\begin{aligned} -\frac{m_{12}}{4}(1-\pi_1)\left(1-\frac{\mu_1}{1+\lambda_1}\right)\tilde{\beta}_1^2 + (1-\pi_1)\frac{\lambda_1}{1+\lambda_1}\mu_1\tilde{\beta}_1\|s\| &\leq \left(\frac{\lambda_1/(1+\lambda_1)\mu_1^2}{m_{12}(1-\mu_1/(1+\lambda_1))}\right)\|s\|^2 \\ &\leq \frac{\mu_1^2}{m_{12}(1-\mu_1)}\|s\|^2 \end{aligned} \quad (33)$$

$$-\frac{m_{13}}{4}(1-\mu_1)\|\tilde{W}_f\|^2 + \mu_1\text{tr}(\tilde{W}_f^T s\phi(x_1, x_2)^T) \leq \frac{\mu_1^2}{m_{13}(1-\mu_1)}\|\phi(x_1, x_2)\|^2\|s\|^2 \quad (34)$$

Substituting (24) and (29)–(34) into (28) gives

$$\begin{aligned} \dot{V}_1 &\leq -\theta_1 s^T s - \frac{m_{11}}{4}(1-\mu_1)\tilde{\alpha}_1^2 - \frac{m_{12}}{4}(1-\pi_1)\left(1-\frac{\mu_1}{1+\lambda_1}\right)\tilde{\beta}_1^2 - \frac{m_{13}}{4}(1-\mu_1)\|\tilde{W}_f\|^2 + \\ &\quad \frac{m_{11}}{2}(1-\mu_1)\alpha_1^2 + \frac{m_{12}}{2}(1-\pi_1)\left(1-\frac{\mu_1}{1+\lambda_1}\right)\beta_1^2 + \frac{m_{13}}{2}(1-\mu_1)\|W_f^*\|^2 \\ &\leq -\rho_1 V_1 + l_1 \end{aligned} \quad (35)$$

where $\rho_1 = \min(2\theta_1, 1/2m_{11}(1-\mu_1), 1/2m_{12}(1-\pi_1)(1-\mu_1/(1+\lambda_1)), 1/2m_{13}(1-\mu_1))$, and

$$l_1 = \frac{m_{11}}{2}(1-\mu_1)\alpha_1^2 + \frac{m_{12}}{2}(1-\pi_1)\left(1-\frac{\mu_1}{1+\lambda_1}\right)\beta_1^2 + \frac{m_{13}}{2}(1-\mu_1)\|W_f^*\|^2 \quad (36)$$

From (35), we obtain

$$0 \leq V_1 \leq \frac{l_1}{\rho_1} + \left(V_1(0) - \frac{l_1}{\rho_1}\right)e^{-\rho_1 t} \quad (37)$$

where $V_1(0)$ is value of V_1 at $t=0$.

It can be seen from (37) that the system was UUD, and a decisive relationship exists between the convergence rate and the parameters m_{11} , m_{12} , and m_{13} . However, (20) is simply a virtual control law,

and $x_{3d} \neq x_3$. Define $e_2 = x_3 - x_{3d}$, and then, the actual differential of V_1 becomes

$$\dot{V}_1 \leq -\rho_1 V_1 + l_1 + s^T B h_\sigma e_2 \quad (38)$$

Step 2. Differentiating e_2 and using (12) gives

$$\dot{e}_2 = W_g^* \phi(x_3) + \varepsilon_g + \gamma p_\sigma v - \dot{x}_{3d} \quad (39)$$

The Lyapunov function is considered as

$$V_2 = V_1 + \frac{1}{2} e_2^T e_2 + \frac{1}{2\gamma_{21}} \tilde{\alpha}_2^2 + \frac{1-\pi_2}{2\gamma_{22}} \tilde{\beta}_2^2 + \frac{1}{2\gamma_{23}} \text{tr}(\tilde{W}_g^T \tilde{W}_g) \quad (40)$$

where $\tilde{\alpha}_2 = \alpha_2 - \hat{\alpha}_2$, $\hat{\alpha}_2$ is the estimate of α_2 ; $\tilde{\beta}_2 = \beta_2 - \hat{\beta}_2$, $\hat{\beta}_2$ is the estimate of $\beta_2 = 1/(1 - \pi_2)$;

$\tilde{W}_g = W_g^* - \hat{W}_g$; and γ_{21} , γ_{22} , and γ_{23} are the update step sizes, respectively.

Differentiating V_2 and using (39) gives

$$\dot{V}_2 \leq -\rho_1 V_1 + l_1 + e_2^T (B h_\sigma s + W_g^* \phi(x_3) + \varepsilon_g + \gamma p_\sigma v - \dot{x}_{3d}) - \gamma_{21}^{-1} \tilde{\alpha}_2 \dot{\tilde{\alpha}}_2 - (1-\pi_2) \gamma_{22}^{-1} \tilde{\beta}_2 \dot{\tilde{\beta}}_2 - \gamma_{23}^{-1} \text{tr}(\tilde{W}_g^T \dot{\tilde{W}}_g) \quad (41)$$

The real control law v is selected as

$$v = \gamma_0^{-1} (-\hat{W}_g \phi(x_3) + \dot{x}_{3d} - k_2 e_2 - \|s^T B\| \text{sgn}(e_2) - \hat{\alpha}_2 \text{sgn}(e_2) - \Lambda_2 \text{sgn}(e_2)) \quad (42)$$

where $\Lambda_2 = -\lambda_2 + \lambda_2 \hat{\beta}_2$ and $\lambda_2 = \|\hat{W}_g \phi(x_3) - \dot{x}_{3d} + k_2 e_2\| + \|s^T B\| + \|\hat{\alpha}_2\| + \kappa_2$, κ_2 is a positive constant.

The adaptive laws are expressed as (43)–(45), respectively, and the control gain k_2 is expressed as (46).

$$\dot{\hat{\alpha}}_2 = \gamma_{21} (1 - \mu_2) (\|e_2\| - m_{21} \hat{\alpha}_2) \quad (43)$$

$$\dot{\hat{\beta}}_2 = \gamma_{22} \left(1 - \frac{\mu_2}{1 + \lambda_2} \right) (\lambda_2 \|e_2\| - m_{22} \hat{\beta}_2) \quad (44)$$

$$\dot{\hat{W}}_g = \gamma_{23} (1 - \mu_2) (e_2 \phi(x_3)^T - m_{23} \hat{W}_g) \quad (45)$$

$$k_2 = \frac{\mu_2^2}{m_{21}(1 - \mu_2)} + \frac{\mu_2^2}{m_{12}(1 - \mu_2)} + \frac{\mu_2^2}{m_{23}(1 - \mu_2)} \|\phi(x_3)\|^2 + \theta_2 \quad (46)$$

where m_{21} , m_{22} , m_{23} , and θ_2 are all positive constants, $\mu_2 = \|\xi_2\|/(\|\xi_2\| + 1)$ is a saturation factor, and ξ_2 is

obtained using the following first-order auxiliary state system:

$$\dot{\xi}_2 = -L_2 \xi_2 + H_2 \Delta v \quad (47)$$

where ξ_2 is an auxiliary state, $\Delta v = v - \text{sat}(v)$ is the saturation nonlinearity of the real control law, and L_2 and H_2 are positive constant matrices.

Substituting (42), (25) and (43) into (41) gives

$$\begin{aligned} \dot{V}_2 \leq & -\rho_1 V_1 - k_2 e_2^T e_2 + l_1 + e_2^T B h_\sigma s - \|s^T B\| \|e_2\| + e^T \left(\tilde{W}_g^* \phi(x_3) - \hat{W}_g \phi(x_3) + \varepsilon_g - \hat{\alpha}_2 \text{sgn}(e_2) - \Lambda_2 \text{sgn}(e_2) \right) - \\ & e_2^T (\gamma_0 - \gamma p_\sigma) v - \gamma_{21}^{-1} \tilde{\alpha}_2 \dot{\hat{\alpha}}_2 - (1 - \pi_2) \gamma_{22}^{-1} \tilde{\beta}_2 \dot{\hat{\beta}}_2 - \gamma_{23}^{-1} \text{tr} \left(\tilde{W}_g^T \dot{\hat{W}}_g \right) \\ \leq & -\rho_1 V_1 - k_2 e_2^T e_2 + l_1 + \mu_2 \text{tr} \left(\tilde{W}_g^T e_2 \phi(x_3) \right) + m_{23} (1 - \mu_2) \text{tr} \left(\tilde{W}_g^T \hat{W}_g \right) + \mu_2 \tilde{\alpha}_2 \|e_2\| + m_{21} (1 - \mu_2) \tilde{\alpha}_2 \hat{\alpha}_2 - \\ & e_2^T (\gamma_0 - \gamma p_\sigma) v - (1 - \pi_2) \gamma_{22}^{-1} \tilde{\beta}_2 \dot{\hat{\beta}}_2 \end{aligned} \quad (48)$$

since

$$-e_2^T (\gamma_0 - \gamma p_\sigma) v \leq \|e_2\| \pi_2 (\lambda_2 \hat{\beta}_2 - \kappa_2) \quad (49)$$

Substituting (44) and (49) into (48) gives

$$\begin{aligned} \dot{V}_2 \leq & -\rho_1 V_1 - k_2 e_2^T e_2 + l_1 + \mu_2 \text{tr} \left(\tilde{W}_g^T e_2 \phi(x_3) \right) + m_{23} (1 - \mu_2) \text{tr} \left(\tilde{W}_g^T \hat{W}_g \right) + \mu_2 \tilde{\alpha}_2 \|e_2\| + m_{21} (1 - \mu_2) \tilde{\alpha}_2 \hat{\alpha}_2 + \\ & (1 - \pi_2) \frac{\lambda_2}{1 + \lambda_2} \mu_2 \tilde{\beta}_2 \|e_2\| + m_{22} (1 - \pi_2) \left(1 - \frac{\mu_2}{1 + \lambda_2} \right) \tilde{\beta}_2 \hat{\beta}_2 \end{aligned} \quad (50)$$

According to *Step 1* and (42), we obtain

$$\dot{V}_2 \leq -\rho_2 V_2 + l_2 \quad (51)$$

where $\rho_2 = \min(\rho_1, 2\theta_2, 1/2m_{21}(1 - \mu_2), 1/2m_{22}(1 - \pi_2)(1 - \mu_2/(1 + \lambda_2)), 1/2m_{23}(1 - \mu_2))$, and

$$l_2 = l_1 + \frac{m_{21}}{2} (1 - \mu_2) \alpha_2^2 + \frac{m_{22}}{2} (1 - \pi_2) \left(1 - \frac{\mu_2}{1 + \lambda_2} \right) \beta_2^2 + \frac{m_{23}}{2} (1 - \mu_2) \|\tilde{W}_g^*\|^2 \quad (52)$$

From (51), it gives as follows

$$0 \leq V_2 \leq \frac{l_2}{\rho_2} + \left(V_2(0) - \frac{l_2}{\rho_2} \right) e^{-\rho_2 t} \quad (53)$$

where $V_2(0)$ is the value of V_2 at $t=0$.

From (53), it can be seen that the trajectory tracking errors are UUD. Through the above analysis, we can get the following conclusion for this study.

Proposition 1. Considering the ROV trajectory tracking system model as (11), virtual control laws as

(20), real control law as (42), adaptive laws as (21)–(23) and (43)–(45), and control gains as (24) and (46), the errors of trajectory tracking control are UUD.

Remark 5. In order to guarantee $\Gamma_1 \geq 0$, the value of $\hat{\beta}_1$ at the initial time is 1. It is similar to $\hat{\beta}_2$.

Remark 6. In the proposed controller, the sign functions $\text{sgn}(s)$ and $\text{sgn}(e)$ are used; they cause chattering in the control system. Therefore, in this study, we use the continuous function $s/(|s| + s_0)$ and $e_2/(|e_2| + e_0)$ instead, where s_0 and e_0 are both small positive scalars. As proved in [50], although it will reduce the robustness of the system, it will not change the tracking errors that are UUD.

Remark 7. The control cycle is usually chosen from 0.1s to 0.5s in the actual ROV control system. Therefore, the proposed method needs less computational effort for online adaptive adjustment.

4. Simulation Studies and Results

4.1 Thruster dynamics

We use a first-order system to approximately describe the dynamic model of the thruster. To verify the feasibility of this approximate method, a simulation is performed.

The parameters of the DC motor driven thruster were set as $C_m=C_e=1/4.32$, $B=0\text{Nm/rad}$, $J=5.06\times 10^{-4}\text{kg.m}^2$, $R_d=0.844\ \Omega$, $L_d=1.07\times 10^{-3}\text{H}$, $K_1=9.6$, and $K_2=0.024\text{V/rad}$. The propeller load torque was $T_p=2.19\times 10^{-5}\omega|\omega|\text{Nm}$. The PI controller gains were designed as $K_P = 45.5$ and $K_I = 22.5$. The control signal of the thruster was an analog voltage from -5.0 V to 5.0 V. Fig. 3 presents the speed control results.

As shown in Fig. 3, by choosing appropriate PI controller gains, a fast response occurred; thus, the actual speed could track the desired speed. However, the current curve indicated a large instantaneous current when the desired speed was changed suddenly. The maximum absolute value of current was 71.5 A. Such a large current has a significant impact on the watertight connector and DC motor, as well as other components. Therefore, the first-order digital filter shown in Fig. 1 was applied, and the time constant was chosen as $T_1 = 0.2$. Fig. 4 presents the speed control results of the thruster with the

first-order digital filter.

Fig. 4 shows that for the same desired speed command from 0s to 4s, the current decreased and the maximum current value was 4.65 A. From 4s to 8s, for the speed step signal, the speed response curve had the characteristics of the first-order system. Subsequently, based on the simulation data of thruster speed control, the first-order system (3) was identified by adaptive learning; $g(\Gamma)$ was identified by the RBF neural networks represented as $g(\Gamma) = \hat{W}\phi(\Gamma)$. The node number of the hidden layer was 16. Based on the K -means clustering algorithm, the centres and bandwidth were chosen. The weights were initialized from zero. The adaptive law for $\hat{W}$ was set as $\dot{\hat{W}} = 0.2\Delta\Gamma\phi(\Gamma)$, where $\Delta\Gamma = \Gamma - \hat{\Gamma}$, Γ is the actual thrust calculated by (2), $a = 1.4 \times 10^{-3}$, and $\hat{\Gamma}$ is the estimated thrust given by model (3). The adaptive law for γ was set as $\dot{\gamma} = 0.1\Delta V$. After identification, a set of data was used for verification.

In Fig. 5, the maximum absolute error is 3.88 N, and the mean square deviation is 1.72 N. The identification results show that there is an estimation error in the RBF neural networks model, but the estimation error is bounded. This is in line with the conditions given in Assumption 4. The simulation results indicated that the first-order system (3) can be used to approximate the thrust model.

4.2 Trajectory tracking control

This section will carry out the verification of the feasibility of the proposed controller. ODIN developed by University of Hawaii is used as a simulation model. ODIN has a near-spherical shape. The weight is 125 kg, and the horizontal diameter and the vertical diameter are 0.63 m and 0.61 m, respectively [51]. The distribution of thrusters is shown in Fig. 6, and the corresponding distribution matrix $\bar{B}$ is defined as (54). The detailed dynamic model can be found in [52].

$$\bar{B} = \begin{bmatrix} \sin\left(\frac{\pi}{4}\right) & -\sin\left(\frac{\pi}{4}\right) & -\sin\left(\frac{\pi}{4}\right) & \sin\left(\frac{\pi}{4}\right) & 0 & 0 & 0 & 0 \\ \sin\left(\frac{\pi}{4}\right) & \sin\left(\frac{\pi}{4}\right) & -\sin\left(\frac{\pi}{4}\right) & -\sin\left(\frac{\pi}{4}\right) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & R \sin\left(\frac{\pi}{4}\right) & R \sin\left(\frac{\pi}{4}\right) & -R \sin\left(\frac{\pi}{4}\right) & -R \sin\left(\frac{\pi}{4}\right) \\ 0 & 0 & 0 & 0 & R \sin\left(\frac{\pi}{4}\right) & -R \sin\left(\frac{\pi}{4}\right) & -R \sin\left(\frac{\pi}{4}\right) & R \sin\left(\frac{\pi}{4}\right) \\ R_z & -R_z & R_z & -R_z & 0 & 0 & 0 & 0 \end{bmatrix} \quad (54)$$

where $R = 0.381$ m and $R_z = 0.508$ m.

In the simulation, we assume that the thrusters of ODIN are the same. The limit thrust value of forward and reverse are 60 N and -60 N, respectively. Moreover, the maximum and minimum control signal values are 5.0 and -5.0 V, respectively. The external disturbances were set as $\Gamma_d = 0.1 \text{rand}(6,1) \cdot x_2$. In real-world applications, the thrust can not be measured directly by sensors when the ROV is working. The benefit from the propeller speed can be obtained, we used the simplified thrust model (2) to calculate the thrust in real time, and therefore, the state value of x_3 could be obtained. In simulation studies, in order to reflect the difference between the actual model and the simplified model, a thrust model as (55) was used [39].

$$\Gamma_i = a\omega_i |\omega_i| - b|\omega_i|v_a, \quad i = 1, \dots, n \quad (55)$$

where $b=0.1$. v_a is the propeller advance speed, which can be obtained through coordinate transformation of the real-time measured body-fixed velocity of ROVs.

In addition, the issue of measurement noise of different sensors was also considered in the simulation. The measurement noise of η was set as $\text{diag}([0.10 \text{ m}, 0.10 \text{ m}, 0.10 \text{ m}, 0.0175 \text{ rad}, 0.0175 \text{ rad}, 0.0175 \text{ rad}]) \cdot \text{rand}(6,1)$, and the measurement noise of v was set as $0.02 \text{rand}(6,1) \cdot v$, where $\text{rand}(\cdot)$ represents the random number between -1 and 1.

In the proposed controller, 16 nodes of the RBF neural networks hidden layer were used, and their centres and bandwidth were chosen by the K -means clustering algorithm. The parameters in the proposed controller were chosen as $\gamma_{11} = \gamma_{12} = \gamma_{13} = 0.2$, and $\gamma_{21} = \gamma_{22} = \gamma_{23} = 0.1$, and $m_{11} = m_{12} = m_{13} =$

$m_{21} = m_{22} = m_{23} = \kappa_1 = \kappa_2 = \varsigma_1 = \varsigma_2 = 0.1$. The initial values are chosen as $\hat{\alpha}_1 = \hat{\alpha}_2 = 0$, and $\hat{\beta}_1 = \hat{\beta}_2 = 1$. $C = \text{diag}([2, 2, 2, 20, 20, 20])$. Figs. 7–9 depict the simulation results.

For the purpose of illustrating the effectiveness of the designed adaptive laws based on the saturation factor, the controller proposed in [8] was used for comparison. In [8], the developed observer-based adaptive controller can get the thruster control signals directly without the thruster dynamics model. The difference from the proposed controller in this paper is that the state observer was adopted and the thruster dynamic and saturation constraints were not considered. Figs. 10–12 illustrate the results of the controller developed in [8].

From 0s to 150s, the desired trajectory changes slowly. There is no thruster saturation constraint except the initial time. In Fig. 7 and Fig. 10, both of two different controllers track the desired trajectory well. In [8], a local recurrent neural network was used for online identification, so that it has better transition characteristics and smaller overshoot when the desired trajectory changes abruptly. This representation can be seen from Fig. 8 and Fig. 11. However, as the controller proposed by [8] has not considered the thruster dynamics, the proposed controller can obtain a better tracking accuracy after trajectory convergence. From 0s to 150s, the mean square deviations of tracking error of trajectory φ obtained by the proposed controller and the comparison method are 0.022 rad and 0.058 rad, respectively. These simulation results are consistent with those given in [8].

After 150s, the curves in Fig. 9 and Fig. 12 demonstrate that the phenomenon of thruster saturation begins to appear as the acceleration of trajectory changes. Because the controller proposed in [8] has not considered the thruster saturation constraints, the convergence rate is very slow and oscillation occurs in Fig. 7 and Fig. 10. If the saturation time is prolonged, it may not guarantee the system stability. However, due to the introduction of saturation factor in the proposed controller, although the trajectory tracking errors will increase after the thrusters are saturation, the tracking errors are UUD.

The simulation results indicated that the proposed control approach was effective. Considering the

thruster dynamics, the proposed controller exhibited stronger performance. Even when the thrusters run in the saturation state, it can guarantee the stability of the trajectory tracking system.

5. Conclusion

An effective adaptive neural sliding mode controller has been presented for trajectory tracking of ROV systems. Considering thruster dynamics, the control signals of the thrusters were designed by the controller directly. Using RBF neural networks for the identification of the unknown nonlinear functions, an adaptive sliding mode controller was proposed using a backstepping technique. By constructing the saturation factor, a new adjustment form of adaptive laws was designed. Simulation results indicated that the controller performed tracking control well even when thruster saturation existed, and the trajectory tracking errors were UUD.

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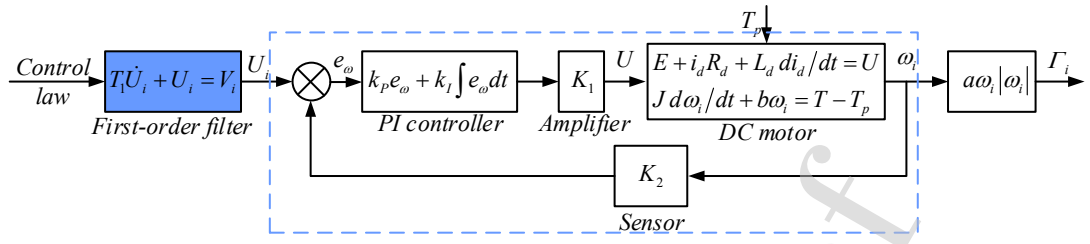


Fig. 1 Thruster's control system.

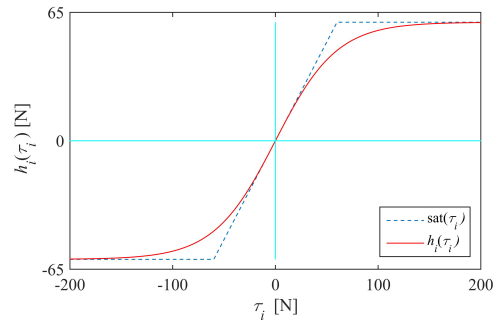


Fig. 2 Smooth saturation function.

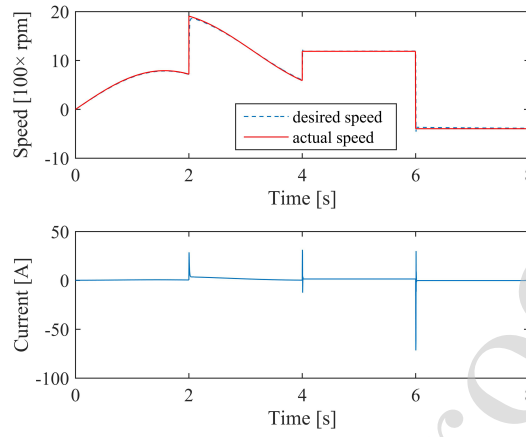


Fig. 3 Thruster's speed control results.

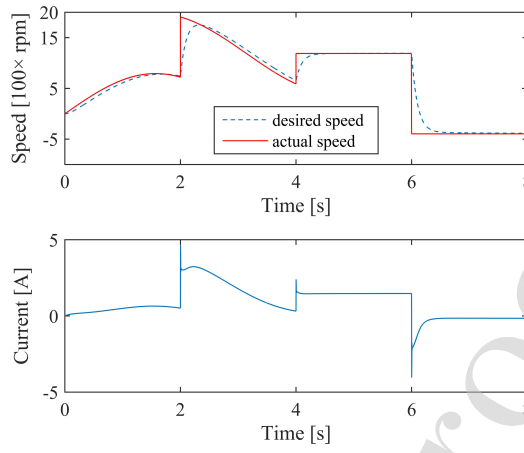


Fig. 4 Thruster's speed control results with first-order filter.

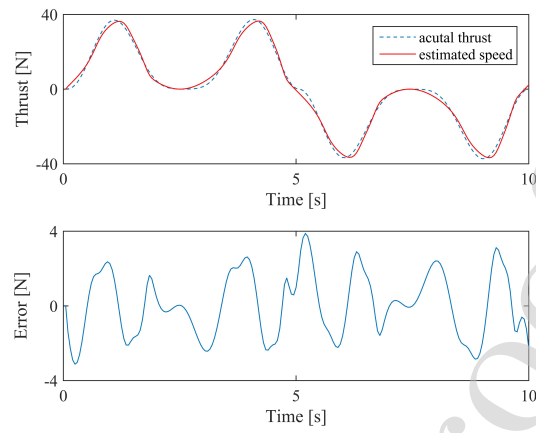


Fig. 5 Identification results of first-order system.

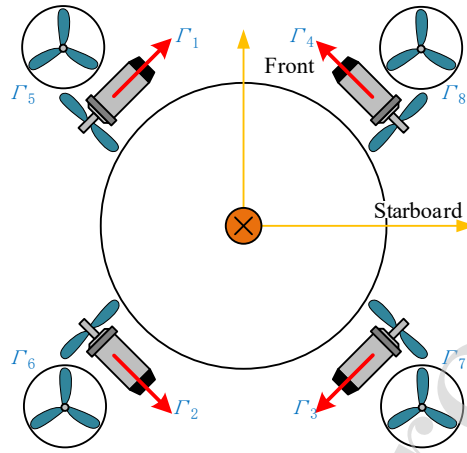


Fig. 6 The distribution of thrusters

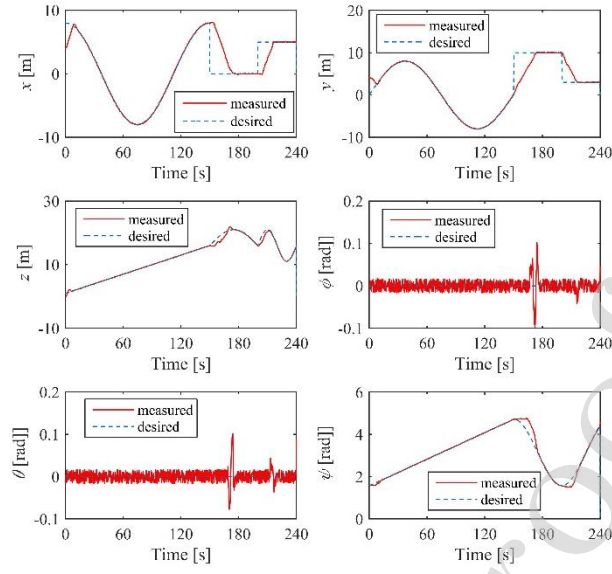


Fig. 7. The trajectory tracking results of the proposed controller

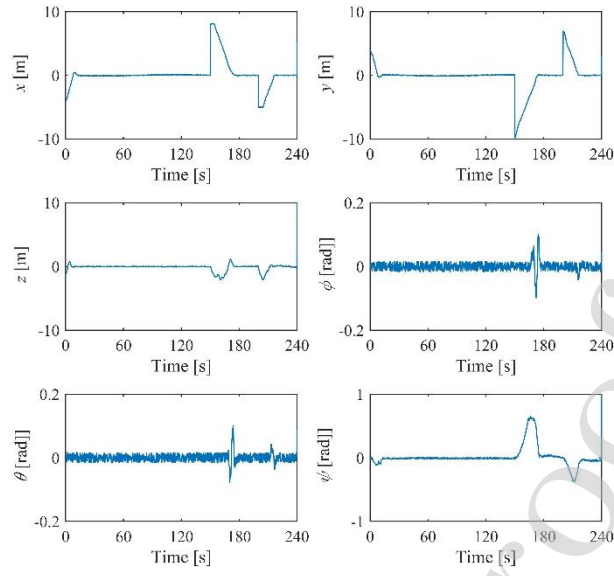


Fig. 8. The trajectory tracking errors of the proposed controller

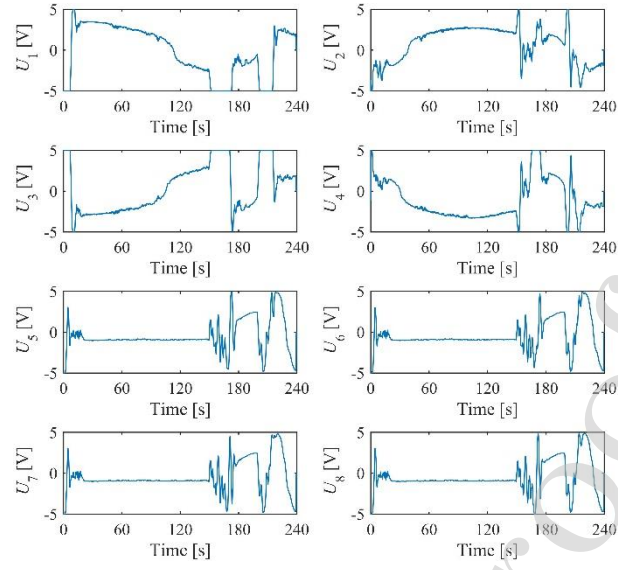


Fig. 9. The thruster control signals of the proposed controller

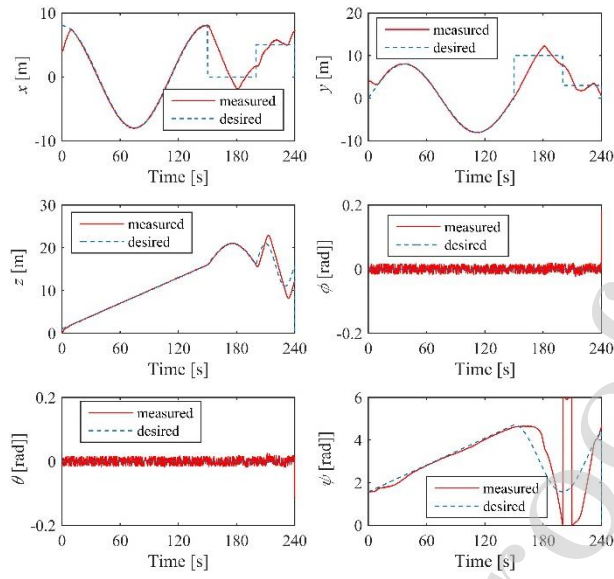


Fig. 10. The trajectory tracking results of the comparison controller

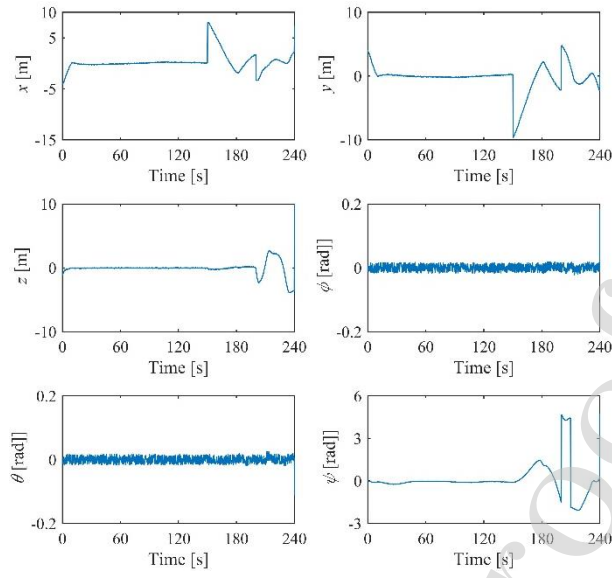


Fig. 11. The trajectory tracking errors of the comparison controller

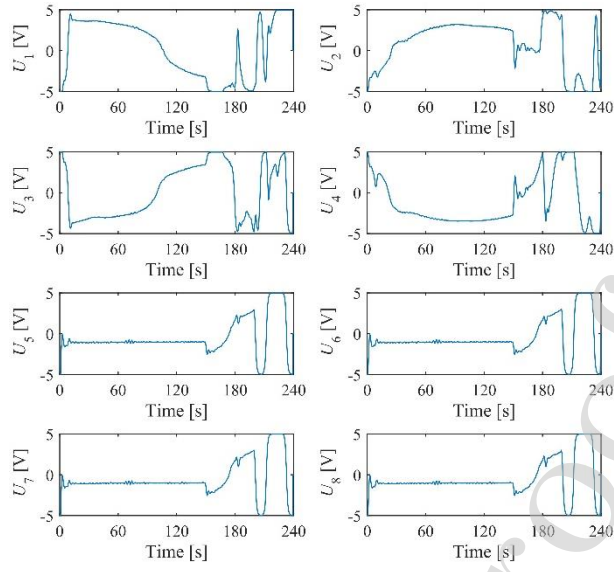


Fig. 12. The thruster control signals of the comparison controller

The thruster dynamics and saturation constraints were considered in controller design.

A third-order state space equation was constructed to describe the dynamic of ROVs.

The unknown dynamics and partially known gain were considered in controller design.

A saturation factor was constructed for the design of the adaptive laws.

Conflict of interest

The authors declared that they have no conflicts of interest to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.